

3.5 Six Sigma for Data Mining

Six Sigma is a popular business management philosophy that focuses on reducing deviations (i.e., sigma) from perfection by rigorously and systematically using proven quality control and business process improvement principles and techniques. This hugely popular management philosophy was first introduced by Motorola in the 1980s in the context of manufacturing process improvement and management. Since then it has been adopted by many companies and organizations in a wide variety of business contexts beyond manufacturing. Ideally, Six Sigma promotes near-zero defects and near-zero tolerance, which can roughly be translated in the business context as error-free perfect business execution. Six Sigma in the business world, also goes by the name DMAIC, involves the following five steps: Define, Measure, Analyze, Improve, and Control.

Because of its success in many other business problems and settings, DMAIC methodology is also applied to data analytics/data science projects. Figure 3.4 shows the DMAIC methodology as a simplified flow diagram. The following sections describe these steps.

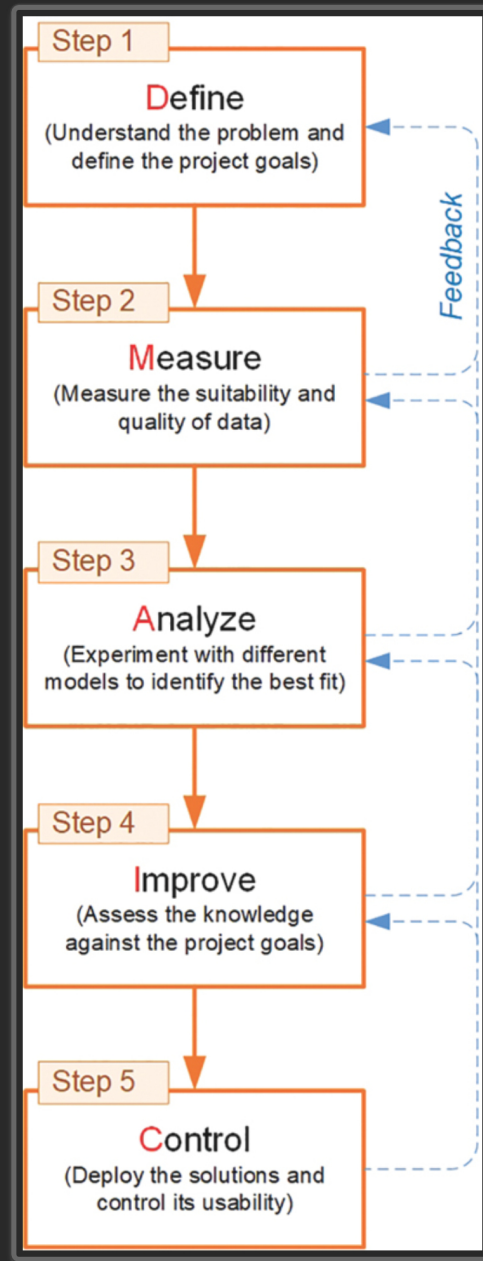


Figure 3.4: Six Sigma DMAIC Methodology

Step 1: Define

The first step in the DMAIC process is about getting the project set up and started. It involves several steps: (1) gaining a thorough understanding of the business needs, (2) identifying the most pressing problem, (3) defining goals and objectives, (4) identifying and defining

the data and other resources needed to investigate the business problem, and (5) developing a detailed project plan. As you may have noticed, there is significant overlap between this step and the first step in the CRISP-DM process.

Step 2: Measure

The second step of the DMAIC process involves assessing the mapping between organizational data repositories and the business problem. Since data mining requires relevant, clean, and usable data, identification and creation of such resources is of critical importance to the success of a data mining project. In this step, the identified data sources are consolidated and transformed into a format that is amenable to machine processing.

Step 3: Analyze

The third step of the DMAIC process is to use a series of data mining techniques to develop models. There are many possible techniques, and most of them are machine learning techniques with many parameters to optimize. There is not a single best technique, so an analyst needs to apply several probable techniques and experiment with them to identify and develop the most appropriate model.

Step 4: Improve

The fourth step of the DMAIC process is to investigate the improvement possibilities. Improvements can be at the technique level or at the business problem level. For instance, if the model results are not satisfactory, other more sophisticated techniques (e.g., ensemble systems) can be used to boost the performance of the models. Also, if the modeling results are not clearly addressing the business problem, the analyst can loop back to previous steps to reexamine and improve the structure of the analysis. The analyst can also further investigate and restate the business problem.

Step 5: Control

The last step of the DMAIC process is to assess the project outcomes, and if they are found to be satisfactory, the models and result are disseminated to decision makers and/or integrated into the existing business intelligence systems for automation.

Six Sigma-based DMAIC methodology resembles the CRISP-DM process. There is no evidence to suggest that one was inspired by the other. However, because these two methodologies involve rather logical and straightforward steps in any business system analysis effort, they need not have inspired one another, and their similarity might simply be coincidental. DMAIC is rarely compared to CRISP-DM because these two methodologies are used to address different types of business applications. Many organizations around the world are using DMAIC and CRISP-DM and their variants to conduct process improvement and data mining projects.